

Artist Network

User Manual

Introduction

Artist Network is a web application that finds the connection path between any two artists using the Last.fm similarity graph. The app allows you to discover how two seemingly unrelated artists are connected through a chain of musically similar artists.

Given a starting artist and a target artist, the application builds a weighted graph where edges represent Last.fm similarity scores, then finds either the shortest path (fewest hops between artists) or the strongest path (highest minimum similarity score along the chain). Each artist along the path is displayed with their top three tracks.

Setup and Installation

Prerequisites

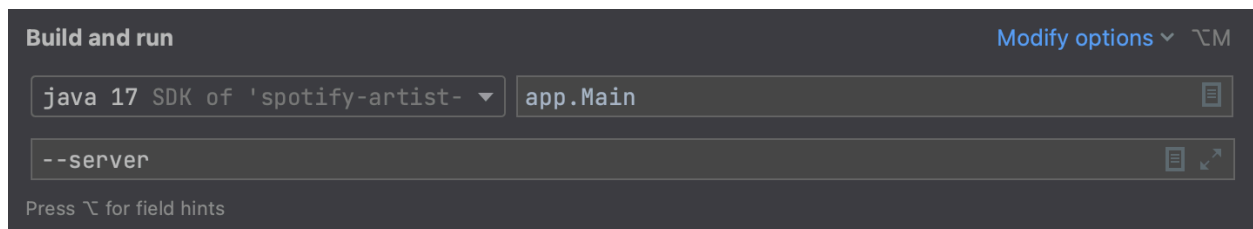
- Java 17 or higher installed on your computer
- A Last.fm API key (free, available at last.fm/api/account/create)
- IntelliJ IDEA or another Java IDE

Installation Steps

1. Clone or download the project repository to your local machine.
2. Open the project folder in IntelliJ IDEA.
3. Add the JSON library to your project dependencies. The library is located at `lib/json.jar` in the project root.
4. Set your Last.fm API key as an environment variable in your run configuration:

```
LASTFM_API_KEY=your_api_key_here
```

5. In your run configuration, set the program argument to:

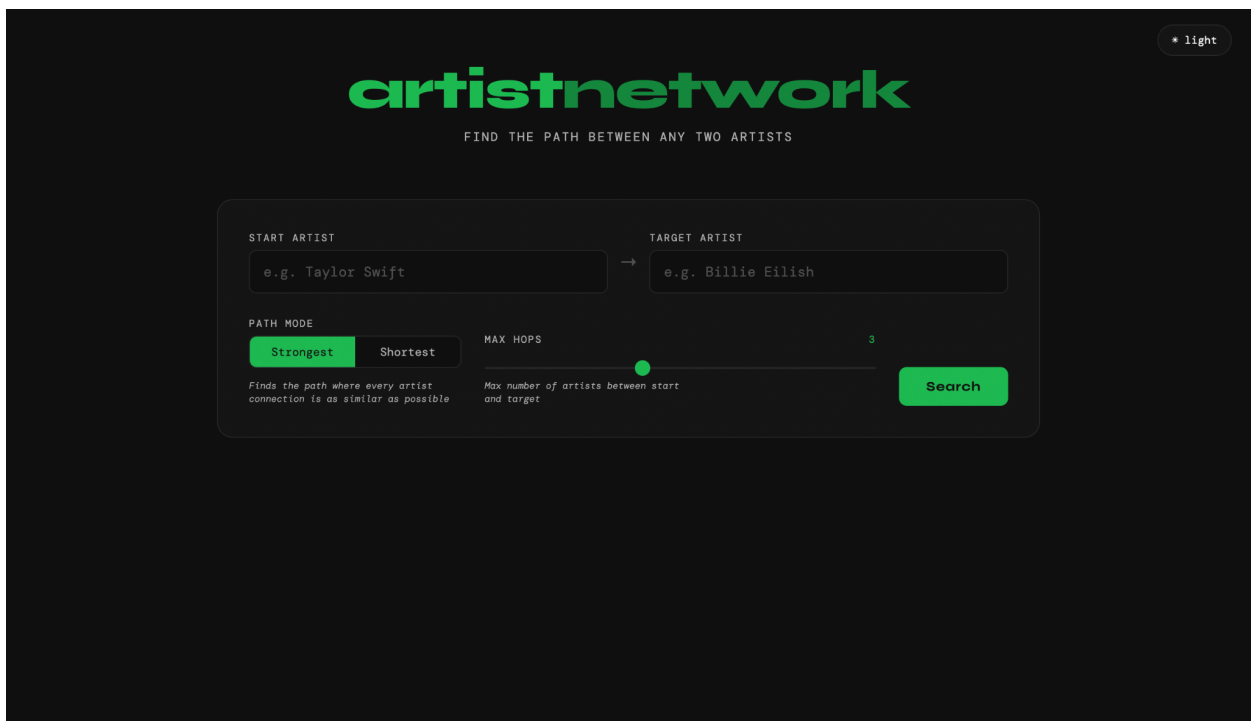


Running the Application

Once your run configuration is set up, click the green play button in IntelliJ to start the server. You will see the following output in the console:

```
=====
Artist Network server running on port 8080
Open http://localhost:8080 in your browser
=====
```

Open your web browser and navigate to:



Using the Application

Searching for a Path

The main interface consists of two text input fields, a path mode toggle, a max hops slider, and a search button.

To find a connection path between two artists:

1. Enter the starting artist name in the Start artist field. For example, type Taylor Swift.
2. Enter the target artist name in the Target artist field. For example, type Billie Eilish.
3. Select your preferred Path mode (see next section).
4. Adjust the Max hops slider to set how many artists can appear in the chain (1 to 6).

5. Click Search or press Enter to find the path.

The screenshot shows a dark-themed application interface. At the top, there are two input fields: 'START ARTIST' containing 'Taylor Swift' and 'TARGET ARTIST' containing 'Billie Eilish', connected by a right-pointing arrow. Below these, the 'PATH MODE' section has two buttons: 'Strongest' (highlighted in green) and 'Shortest'. To the right of these buttons is a 'MAX HOPS' slider with a green dot and the number '3' at its end. Below the slider is a green 'Search' button. Small text descriptions are present: 'Finds the path where every artist connection is as similar as possible' under the 'Strongest' button, and 'Max number of artists between start and target' under the 'MAX HOPS' slider.

Path Modes

The application offers two different ways to find paths between artists:

Strongest Path mode finds the chain where every connection between adjacent artists is as similar as possible. This may result in a longer path but ensures each step represents a strong musical similarity. The path score in this mode equals the strength of the weakest link in the chain.

Shortest Path mode finds the chain with the fewest hops between artists, regardless of how strong each individual connection is. This produces faster results but the path may include weaker similarity connections.

This block contains two side-by-side panels illustrating the path modes. The left panel, titled 'PATH MODE', shows the 'Strongest' button highlighted in green and the 'Shortest' button in grey. Below the buttons, it says 'Finds the path where every artist connection is as similar as possible'. The right panel, also titled 'PATH MODE', shows the 'Strongest' button in grey and the 'Shortest' button highlighted in green. Below the buttons, it says 'Finds the path with the fewest hops between artists'.

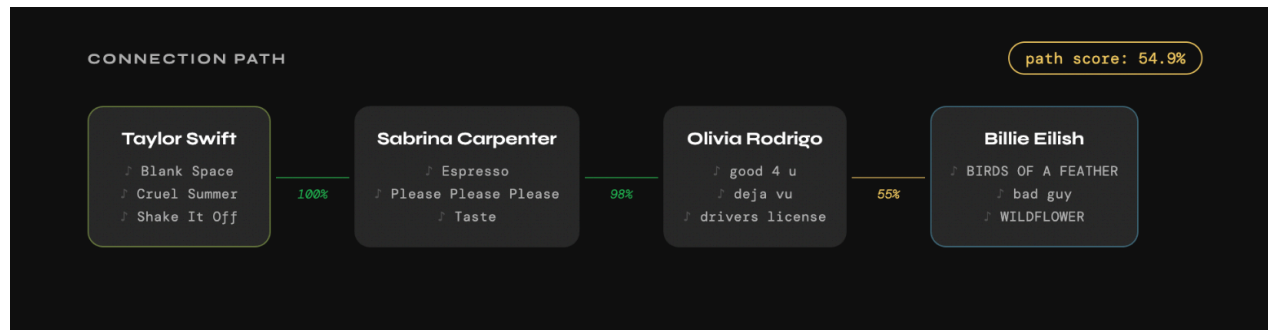
Max Hops Slider

The max hops slider controls how many intermediate artists can appear in the connection path. A lower value forces the algorithm to find a more direct connection, while a higher value allows the algorithm to search through more artists to find a connection.

If no path is found at a low hop count, try increasing the slider to allow the algorithm to search further. The default value is 3 hops.

Reading the Results

After clicking Search, the application displays the connection path as a series of cards connected by lines. Each card represents an artist in the chain and shows their top three tracks underneath their name.



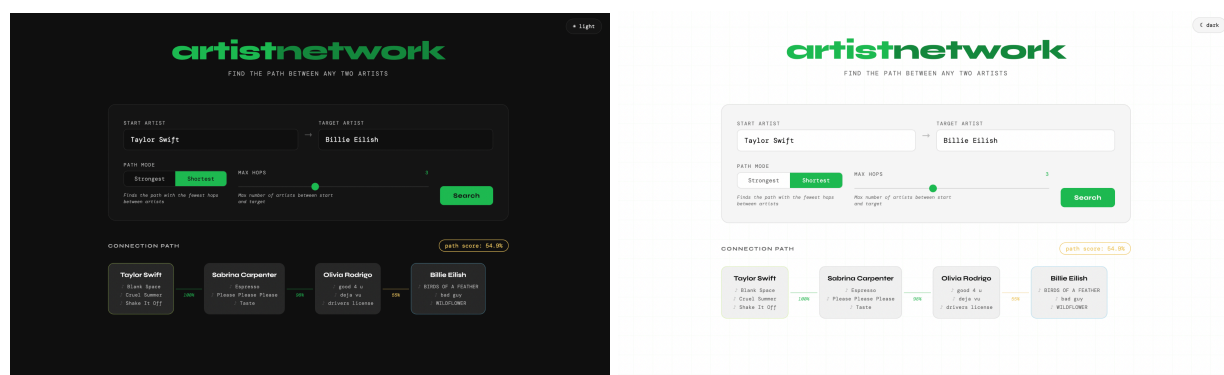
The lines connecting the cards display similarity percentages. These percentages indicate how similar each pair of adjacent artists is according to Last.fm. The colors of these connections also indicate strength:

- **Green connections** (70 percent or higher) indicate strong similarity
- **Yellow connections** (40 to 69 percent) indicate moderate similarity
- **Red connections** (below 40 percent) indicate weak similarity

In the upper right of the results section, the overall path score is displayed. This represents the strength of the weakest link in the chain.

Light and Dark Mode

The application supports both light and dark color schemes. To toggle between modes, click the button in the top right corner of the screen labeled with a sun (light) or moon (dark) icon.

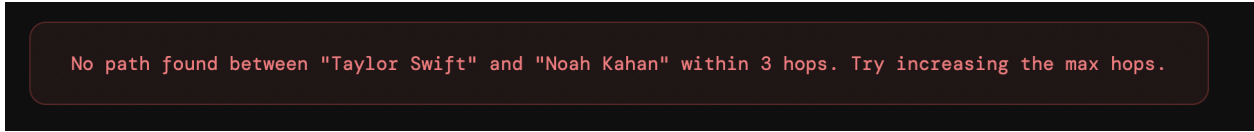


Error Messages

If the application cannot find a path or encounters an issue, an error message will be displayed.

No Path Found

If no path can be found between the two artists within the selected number of hops, the application will display a message such as:

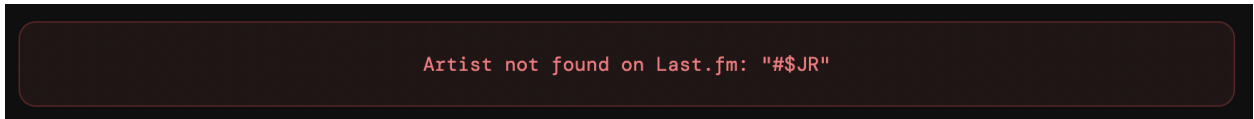


No path found between "Taylor Swift" and "Noah Kahan" within 3 hops. Try increasing the max hops.

In this case, increase the max hops slider and try again.

Artist Not Found

If the artist name does not match any artist in the Last.fm database, the application will display an error indicating which artist could not be found. Make sure the artist name is spelled correctly and matches the official spelling on [Last.fm](https://www.last.fm).



Artist not found on Last.fm: "\$JR"

Server Connection Errors

If the application cannot reach the server, it will display an error asking you to confirm that the Java server is running. Return to IntelliJ and verify that the server is still active in the console.

Tips for Best Results

- Use exact artist names as they appear on Last.fm. Capitalization matters less than spelling.
- The first search after starting the server may take longer because the graph is being built. Subsequent searches will be faster as the graph is cached.
- If you receive a no path found error, try increasing the max hops slider before changing your artist selections.
- For artists from very different genres, you will likely need a higher hop count to find a connection.
- Use the Strongest path mode when you want to discover artists that are genuinely similar to your starting artist. Use Shortest when you just want to see the most direct connection possible.

Troubleshooting

If you encounter issues running the application, verify the following:

- Your LASTFM_API_KEY environment variable is set correctly in your run configuration

- Your Java version is 17 or higher
- The json.jar dependency is included in your project structure
- The --server program argument is set in your run configuration
- The src/resources folder is marked as a Resources Root in IntelliJ
- Port 8080 is not being used by another application on your computer